

Eunwoo Kim

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EDUCATION

M.S.	2026	Psychology: Quantitative Psychology (PI: Inhan Kang)	Yonsei University
B.S.	2024	Computer Science and Engineering	Seoul National University
B.A.	2024	Psychology	Seoul National University

RESEARCH INTERESTS

Cognitive Modeling (Reinforcement learning, Sequential sampling models), Model-based Cognitive Neuroscience, Machine Learning

WORKING PROJECTS

1.	Development and Validation of a Reinforcement Learning Sequential Sampling Model for Addictive Behavior with Inhan Kang	<i>Submitted</i>
2.	Joint Modeling of Behavioral and Neural Data via Lasso Factor-Analytic Neural Diffusion Decision Models: Reward-Modulated Recognition Memory with Seoyoung Choi, Sungwoo Ahn, Sanghoon Han, Inhan Kang	<i>Data analysis in progress</i>
3.	Predicting Fertility Desires Among Engaged Couples: An Interpretable Machine Learning Approach Using Dyadic Data with Hyounghbin Moon, Young-Hoon Kim, Inhan Kang	<i>Manuscript in preparation</i>

CONFERENCE PRESENTATIONS

1.	Development and Validation of a Reinforcement Learning Sequential Sampling Model for Addictive Behavior	
•	International Meeting of the Psychometric Society (IMPS), Oral presentation	Jul. 2026
•	Conference of the Korean Society of Psychological Measurement and Assessment, Oral presentation	Oct. 2025
•	Annual Conference of the Korean Psychological Association, Poster presentation	Aug. 2025

RESEARCH EXPERIENCES

Research Assistant Feb.–Aug. 2024

Connectome Laboratory (PI: Jiook Cha), *Seoul National University*

- Led an independent project applying Generalized Random Forests to estimate heterogeneous treatment effects of COVID-19-related variables on well-being
- Collaborated with Center of Happiness Studies, Seoul National University, which provided the dataset, and jointly interpreted the results

Research Assistant Jan.–Dec. 2023

Quantitative Research Collaboratory Laboratory (PI: Amanda K. Montoya),
University of California, Los Angeles

- Conducted literature reviews for projects on tree-based ML methods in psychology, SPSS/SAS mediation/moderation macro, and a registered-reports survey study
- Assisted with survey administration for the registered-reports project
- Served as an independent analyst in a many-analysts-style project examining variability across independent analyses of the same dataset

Research Assistant

Aug.–Dec. 2022

Computational Clinical Science Laboratory (PI: Woo-Young Ahn), *Seoul National University*

- Contributed data visualization code to a project analyzing orthogonal go/no-go task data
- Participated in a weekly literature study group on decision-making and computational methods

AWARDS

- Future Psychologist Award, Korean Psychological Association (2025)
- Summa Cum Laude, Department of Psychology, Seoul National University (2024)

TEACHING EXPERIENCES**Teaching Assistant**

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| • Bayesian Statistics for Psychology (Graduate) | Fall | 2025 |
| • Psychological Statistics (Undergraduate) | Spring | 2025 |
| • Data Science for Analysis of Human Behavior and Psychology (Undergraduate) | Fall | 2024 |

SELECTED COURSEWORK**Psychology/Neuroscience** – Advanced cognitive science, Seminar in Quantitative Psychology, Seminar in Experimental Psychology, Psychometrics, Neuroimaging, Neuroscience**Statistics** – Mathematical statistics (graduate), Linear model (graduate), Machine learning theory and application, Generalized mixed models for data science, Structural equation modeling, Linear algebra**SKILL SET****Statistical Software** – R, Matlab / **Bayesian** – Stan·Rstan, hBayesDM / **Programming** – Python, C, Java**REFERENCES****Inhan Kang** (E-mail: qpsy@yonsei.ac.kr)

Professor, Department of Psychology, Yonsei University

Sanghoon Han (E-mail: sanghoon.han@yonsei.ac.kr)

Professor, Department of Psychology, Yonsei University

Hanjoe Kim (E-mail: hanjoekim@yonsei.ac.kr)

Professor, Department of Psychology, Yonsei University

APPENDIX: SUMMARY OF RESEARCH

WORKING PROJECTS

Development and Validation of a Reinforcement Learning Sequential Sampling Model for Addictive Behavior

Eunwoo Kim, Inhan Kang
(Submitted, *Psychonomic Bulletin & Review*)

Abstract : Behavioral addiction is characterized by a persistent preference for immediate rewards over long-term outcomes, reflecting a breakdown in value-based decision-making. To elucidate the mechanism underlying this maladaptive pattern, researchers have applied reinforcement learning (RL) models to the Iowa Gambling Task (IGT). However, two limitations remain. First, parameter recovery has received limited scrutiny, raising concerns about the reliability of individual-level estimates. Second, these models rely solely on choice data and ignore other informative modalities such as response times (RTs), although RTs can be collected easily and provide distinct information from choice. To address these limitations, we integrated an RL model for the IGT with an evidence-accumulation structure of sequential sampling models (SSMs), resulting in a joint model of IGT choices and RTs. The RL component updates subjective deck values based on trial-by-trial feedback, and the SSM component links these values to within-trial decision dynamics. We validate the proposed model with two approaches. First, a simulation study shows that all seven individual-level parameters are recovered accurately. Second, absolute model fits to IGT data from 20 older adults demonstrate that the model reproduces card-wise choice trajectories, win-stay/lose-shift patterns, and empirical RT distributions. The results also show the joint model performs as well as or better than several choice-only RL models for the IGT, in both simulation and empirical application. Overall, our modeling approach provides a unified account of choice and RT in the IGT while yielding reliable individual-level estimates for investigating value-based decision-making processes.